

MAMMOTH YOSEMITE, CA, USA

WMO: 723894

Lat: 37.633N

Lon: 118.850W

Elev: 7128

StdP: 11.28

Time Zone: -8.00 (NAP)

Period: 08-19

WBAN: 03181

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.2	5.1	-11.2	3.9	24.7	-7.2	4.9	25.1	35.2	43.3	28.9	42.0	2.5	350	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.1	86.3	54.2	84.3	53.5	82.2	52.6	57.3	77.3	56.0	77.1	54.8	76.7	11.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.8	72.0	61.5	47.9	64.5	61.6	45.3	58.3	60.8	27.9	77.1	26.8	76.4	25.9	76.8	62.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	23.0	19.2	DB	-12.7	90.1	7.2	2.4	-17.9	91.8	-22.1	93.2	-26.1	94.5	-31.4	96.2	(4)
(5)			WB	-13.2	60.4	7.1	1.1	-18.3	61.2	-22.4	61.9	-26.4	62.5	-31.6	63.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	28.9	30.5	36.0	42.9	49.1	59.2	64.5	62.7	56.7	46.3	36.4	28.4	(6)	
	DBStd	14.60	9.24	9.62	7.60	7.37	6.46	6.23	4.23	3.86	5.33	5.84	7.95	8.80	(7)	
	HDD50	3195	654	547	435	226	94	8	0	0	14	139	408	670	(8)	
	HDD65	7301	1119	966	900	664	492	190	59	90	251	578	857	1135	(9)	
	CDD50	1447	0	1	0	12	67	282	449	394	214	26	2	0	(10)	
	CDD65	78	0	0	0	0	0	15	43	19	1	0	0	0	(11)	
	CDH74	4287	0	0	0	2	42	830	1636	1325	429	23	0	0	(12)	
	CDH80	1032	0	0	0	0	1	202	473	314	42	0	0	0	(13)	
(14) Wind	WSAvg	6.6	4.9	6.4	7.1	8.5	8.3	8.0	6.5	6.2	6.1	5.8	5.5	5.7	(14)	
(15) Precipitation	PrecAvg	13.6	2.2	2.1	1.7	1.0	0.8	0.4	0.5	0.4	0.4	0.8	1.0	2.0	(15)	
	PrecMax	20.7	6.0	7.0	6.8	3.0	2.0	1.7	1.1	0.8	1.4	2.9	3.5	7.4	(16)	
	PrecMin	6.9	0.2	0.3	0.1	0.2	0.2	0.0	0.1	0.1	0.0	0.0	0.1	0.2	(17)	
	PrecStd	3.4	1.7	1.3	1.6	0.6	0.5	0.4	0.3	0.2	0.3	0.7	0.8	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.3	63.1	66.1	72.5	78.6	88.5	89.9	88.2	83.6	77.1	68.4	59.3	(19)	
		MCWB	39.3	41.0	43.0	47.1	49.2	55.1	55.3	54.4	51.5	47.9	44.1	38.9	(20)	
	2%	DB	54.4	58.8	61.4	68.4	74.6	85.0	87.1	85.6	81.2	72.5	63.9	54.8	(21)	
		MCWB	36.9	38.8	39.7	44.6	47.7	53.4	54.7	53.5	51.2	46.3	42.1	36.7	(22)	
	5%	DB	50.0	54.2	57.4	65.6	71.7	82.0	84.8	83.7	79.0	68.5	59.8	50.2	(23)	
		MCWB	35.6	36.8	38.4	43.1	46.8	51.9	54.2	52.9	50.3	44.9	40.0	35.6	(24)	
	10%	DB	45.0	48.1	53.9	61.7	67.9	79.2	82.3	81.4	76.6	64.5	54.8	45.8	(25)	
		MCWB	33.7	34.8	37.5	41.5	45.3	51.0	53.5	52.1	49.6	43.2	37.9	33.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.7	42.1	43.5	48.6	51.0	57.1	59.5	58.2	56.3	50.7	45.7	41.7	(27)	
		MCDB	54.4	59.1	63.0	68.3	73.5	81.9	76.7	77.4	73.0	66.2	64.4	53.2	(28)	
	2%	WB	38.3	39.8	41.2	46.1	49.3	55.2	58.0	56.4	53.9	48.6	43.2	39.1	(29)	
		MCDB	49.7	55.5	58.7	65.1	70.2	80.1	76.9	76.9	74.7	66.5	60.6	49.9	(30)	
	5%	WB	36.7	37.4	39.6	44.0	47.9	53.5	56.7	55.0	52.2	46.4	41.2	36.9	(31)	
		MCDB	46.9	51.3	55.8	63.2	67.6	77.9	76.9	76.4	74.1	64.1	57.5	48.4	(32)	
	10%	WB	34.4	35.3	37.8	42.2	46.3	52.0	55.5	53.8	50.7	44.6	38.9	34.4	(33)	
		MCDB	43.3	47.5	51.2	60.0	65.0	75.7	76.8	76.4	73.2	61.5	53.7	45.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	28.1	27.0	27.9	29.9	30.2	34.5	36.1	37.7	37.0	34.9	32.9	28.2	(35)	
		MCDBR	35.6	35.2	36.1	37.0	37.2	39.3	38.2	40.6	40.7	41.5	40.5	35.2	(36)	
	5% WB	MCWBR	20.9	18.8	18.2	17.5	16.7	15.8	14.5	16.2	17.6	20.0	21.5	20.2	(37)	
		MCWBR	29.0	32.2	33.0	34.6	34.7	36.4	33.4	35.1	35.7	35.9	37.6	30.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.245	0.260	0.261	0.298	0.303	0.298	0.314	0.298	0.282	0.269	0.244	0.242	(40)	
		taud	2.427	2.383	2.436	2.402	2.466	2.503	2.503	2.532	2.582	2.592	2.599	2.485	(41)	
	Ebn at Noon		305	312	321	312	310	310	304	308	310	306	305	299	(42)	
		Edh at Noon	26	31	33	36	35	34	34	32	29	25	22	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		830	1112	1528	1965	2269	2627	2485	2318	1954	1427	983	736	(44)	
	RadStd		81	117	117	126	183	167	146	118	76	100	56	84	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (18 neighbors)	+0.97	N/A	N/A	+0.79	N/A	N/A	-85	-203	+236	+119

Nomenclature: See separate page